

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
		(iii)	Using equation for coefficient of restitution $e = \sqrt{\frac{h}{H}}$ (1) Coefficient of restitution = 0.39 (1) The same value for both surfaces (1) If no square root taken award a max of 2 marks			3	3	2	
		(iv)	Recall $T = I\alpha$; (1) Torque = 6.2 [N m] (1) Alternative: Recall: torque = $\frac{\text{change in angular momentum}}{\text{time}}$ (1) Torque = 6.2 [N m] (1)	1	1		2		
		(v)	Either of the vertical components calculated correctly $12\sin 32^\circ = 6.4 \text{ [m s}^{-1}\text{]}$ or $12\cos 32^\circ = 10.2 \text{ [m s}^{-1}\text{]}$ (1) Substitute values into $v^2 = u^2 + 2ax$ or $x = ut + \frac{1}{2}at^2$ (1) Time to reach height = 0.65 [s] or height $x = \frac{6.4^2}{2 \times 9.81}$ or $t = \frac{6.2}{10\cos 32}$ [= 0.61 s] (1) [from level of the thrower] Maximum height reached by the ball = 2.1 [m] [from level of the thrower] or height reached = 2.05 [m] (1) Overall, height = $\frac{2.1}{2.05} + 2.0$ [= 4.1 m] so jumper will catch the ball (1)	1 1	1 1 1		5	5	
			Question 9 total	6	9	5	20	10	0